

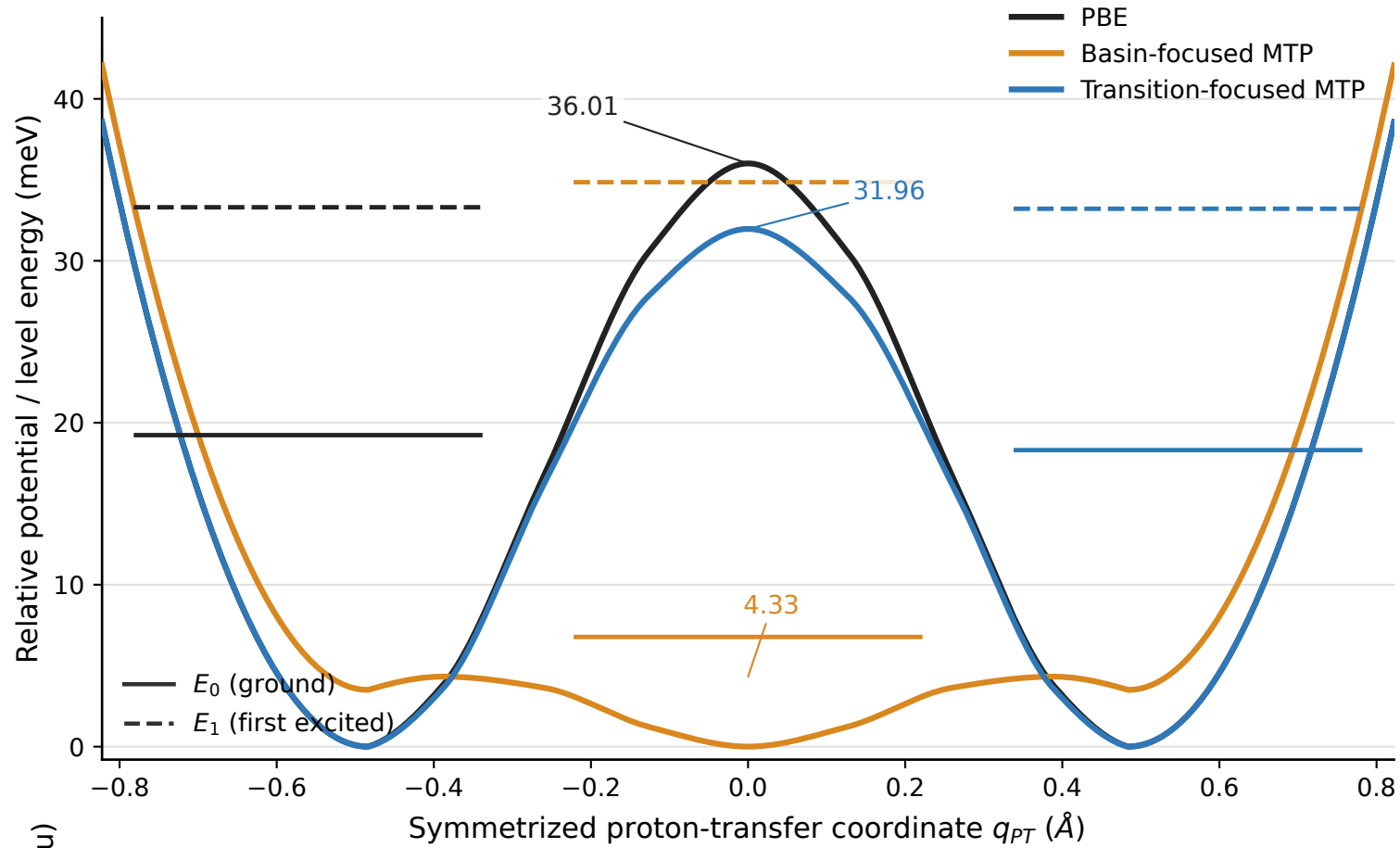
Frozen-path 1D quantum-level audit of proton transfer in malonaldehyde

Transition-focused equal-budget sampling better preserves the PBE-derived low-level spectrum on the frozen PBE path.

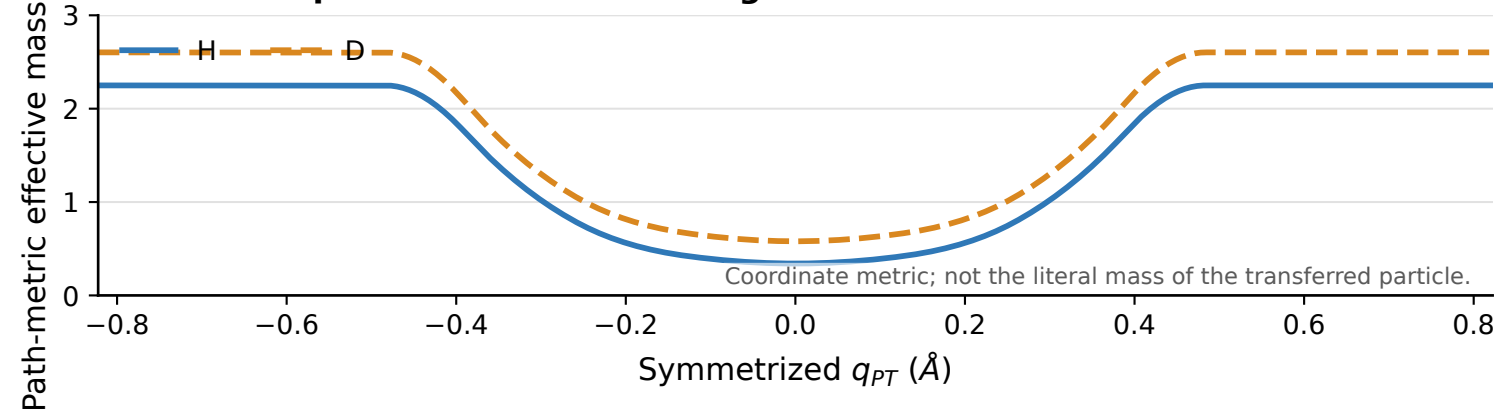
Frozen-path one-dimensional spectral diagnostic; not an experimental tunneling-rate prediction.

a

Frozen-path 1D potentials and two lowest H levels

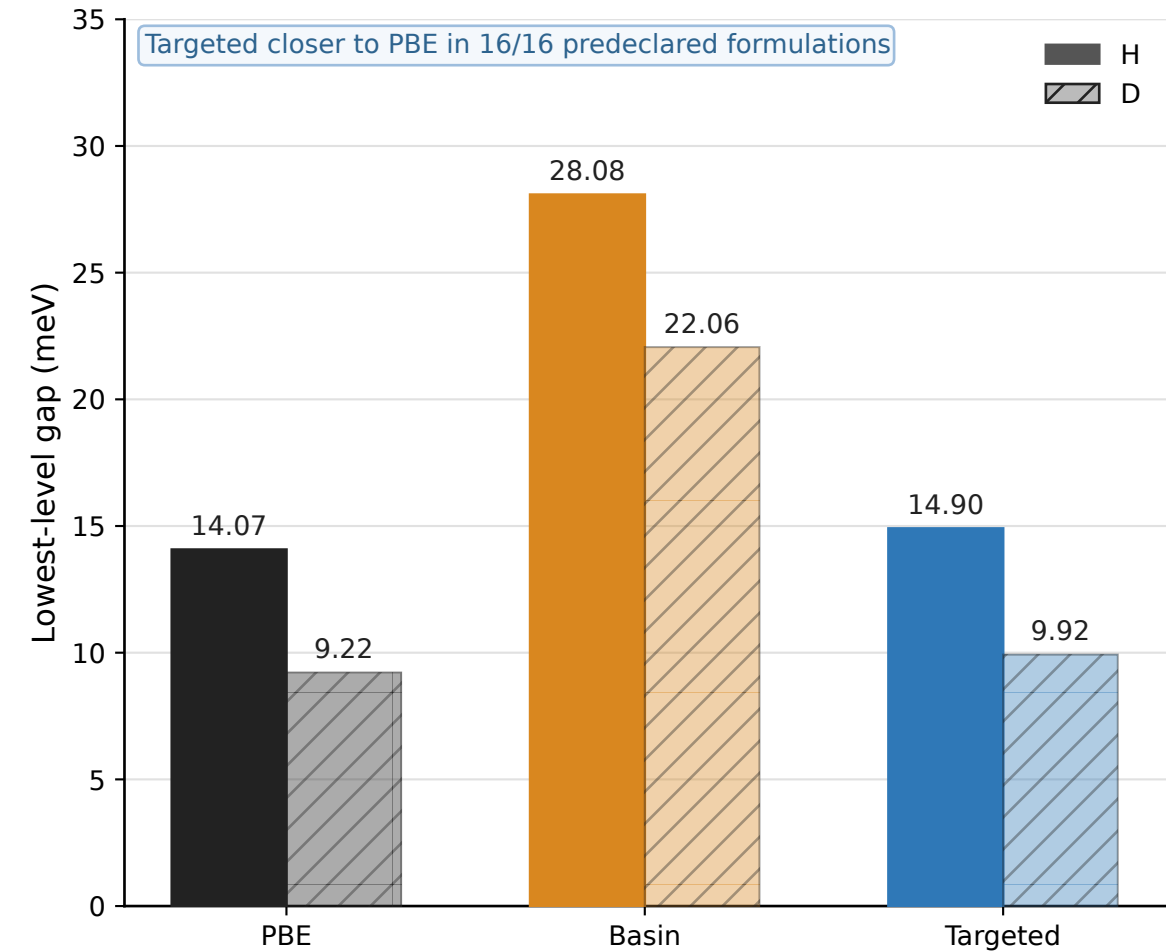


Frozen PBE-path effective-mass diagnostic



b

PBE-derived H/D lowest-level gaps



Primary H-level classification

PBE

sub-barrier doublet

Basin

no low sub-barrier doublet

Targeted

only E_0 below barrier